

STEEL STRUCTURE CONVERSION

Tekla IFC → E3D Structure Model

Parses Tekla IFC via ifcopenshell, extracts extrusion profiles / world placement, converts IfcBeam/Plate to E3D SCTN/PANE.

Key Purpose

Why this program

1. Re-model the steel structure in E3D without redrawing

- Parses the Tekla IFC, extracts each member's profile and world placement, and writes a PML macro that builds SCTN and PANE (plate) elements

→ **Effect:** A full steel frame is rebuilt in E3D in minutes with no manual re-modelling and no transcription error

2. Automatic section catalog mapping

- Unique IFC profile names are guessed into E3D SPREF and written to a mapping CSV the user can correct, then re-run

→ **Effect:** Members land on the right catalog sections; blanks or mismatches are fixed once in the CSV instead of member by member

3. Correct orientation and level structure

- Derives BANG from the web direction, sets JUSLINE (horizontal=TOS, vertical=CENT), and splits members into STRU/SUBS by plan elevation

→ **Effect:** Members sit at the right rotation and justification, grouped by floor level, so the model needs

Required inputs : Tekla Structures IFC file (.ifc, IFC2X3 AISC Steel Detailing export)

bolts as SOLIDS

Main Window – Item Guide

The screenshot shows the 'Tekla IFC → E3D (SCTN/PANE) Converter' application window. The interface includes a title bar, a menu bar, and several sections. The 'Select Tekla IFC File' section has a text field (1) and a 'Browse...' button (2). The 'Option' section contains 'Elev Tolerance(mm):' (3), 'Beam Jusline:' (4), 'Elev Prefix:' (5), and 'Bolt/Nut Create:' (6). The 'Progress / Result' section features a large text area (7). The 'Output Files (double-click → UltraEdit)' section has a text field (8). At the bottom, there are 'IFC Preview' (9) and 'Convert' (10) buttons.

1. Select Tekla IFC file (path field)
2. Browse for IFC file
3. Elev Tolerance (mm) – level grouping
4. Beam Jusline (TOS / CENT)
5. Elev Prefix (EL / FL / GL) for STRU names
6. Bolt/Nut Create (Off = skip washers/anchors)
7. Progress / Result log
8. Output Files (double-click to open)
9. IFC Preview (inspect before converting)
10. Convert – generate E3D macro & mapping

Overview

What it does

1. Parses Tekla IFC via ifcopenshell
2. Extracts extrusion profiles
3. World placement
4. Converts IfcBeam/Plate to E3D SCTN/PANE

Category: Steel Structure Conversion

Folder: STEEL_STRUCTURE/TEKLATOE3D

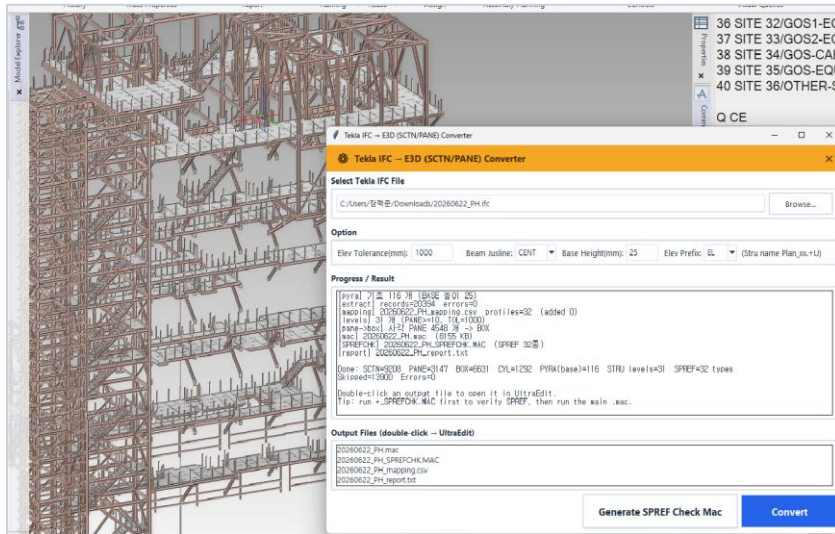
Main scripts: teklaifctoe3d.py

Scripts / Status: 2 scripts · Operational

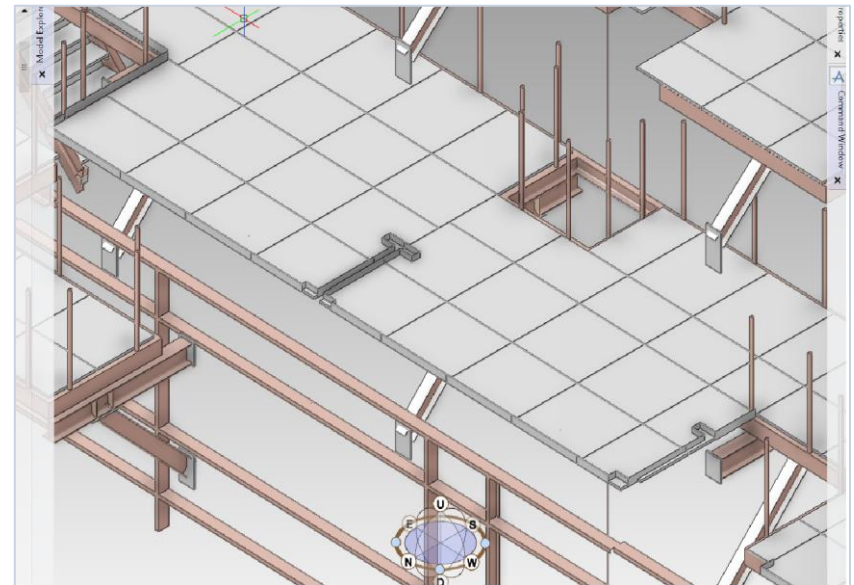
Tags: IFC · ifcopenshell · Tekla · SCTN

Output Samples

Output Samples (1/3)



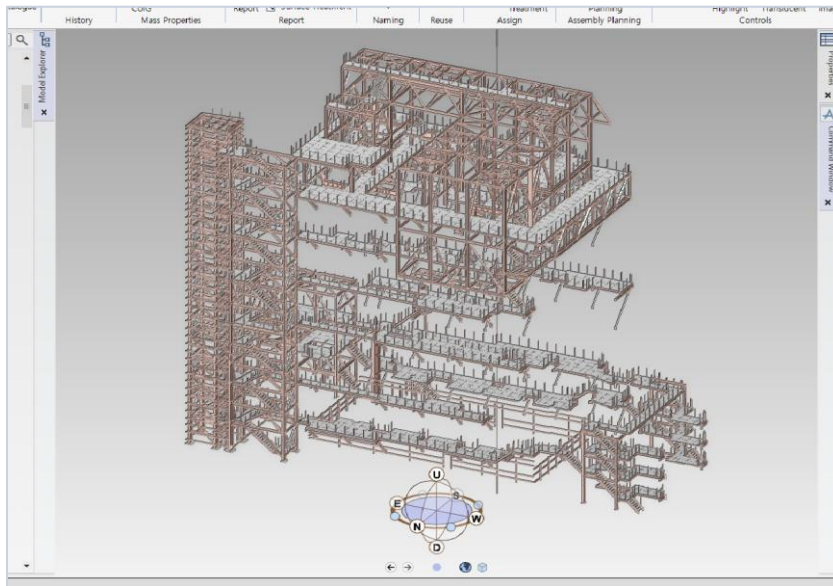
output0



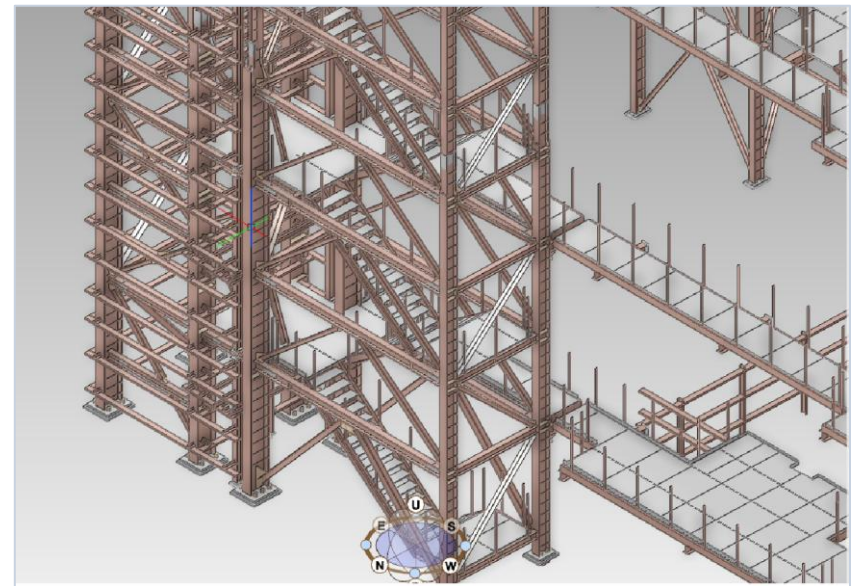
output02

Output Samples

Output Samples (2/3)



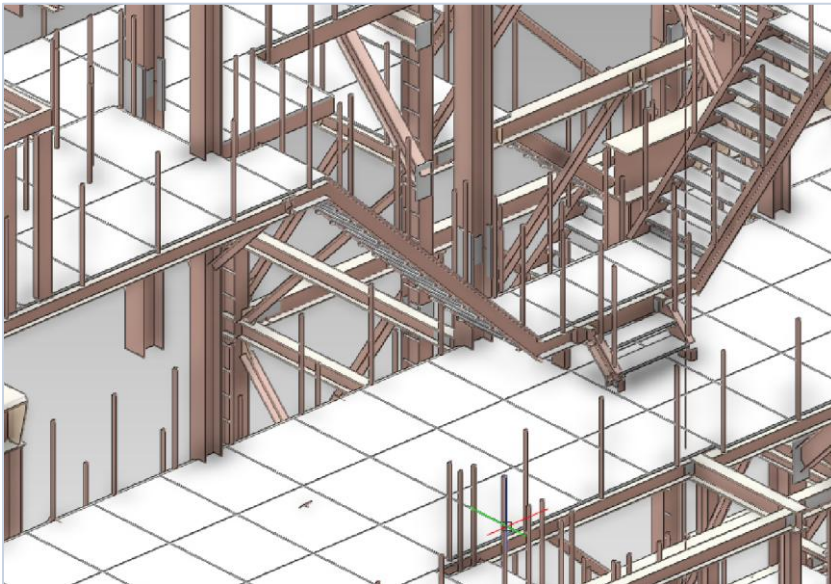
output1



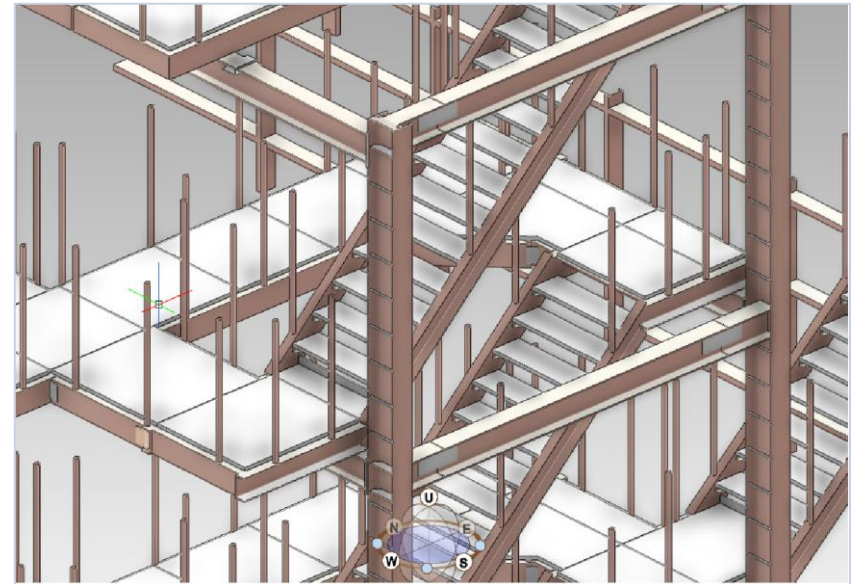
output3

Output Samples

Output Samples (3/3)



output4



output5

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Steel Structure Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: teklaifctoe3d.py.
- Output samples and the program window above reflect the current build.

Thank You

Tekla IFC → E3D Structure Model · Plant Engineering Total Service System